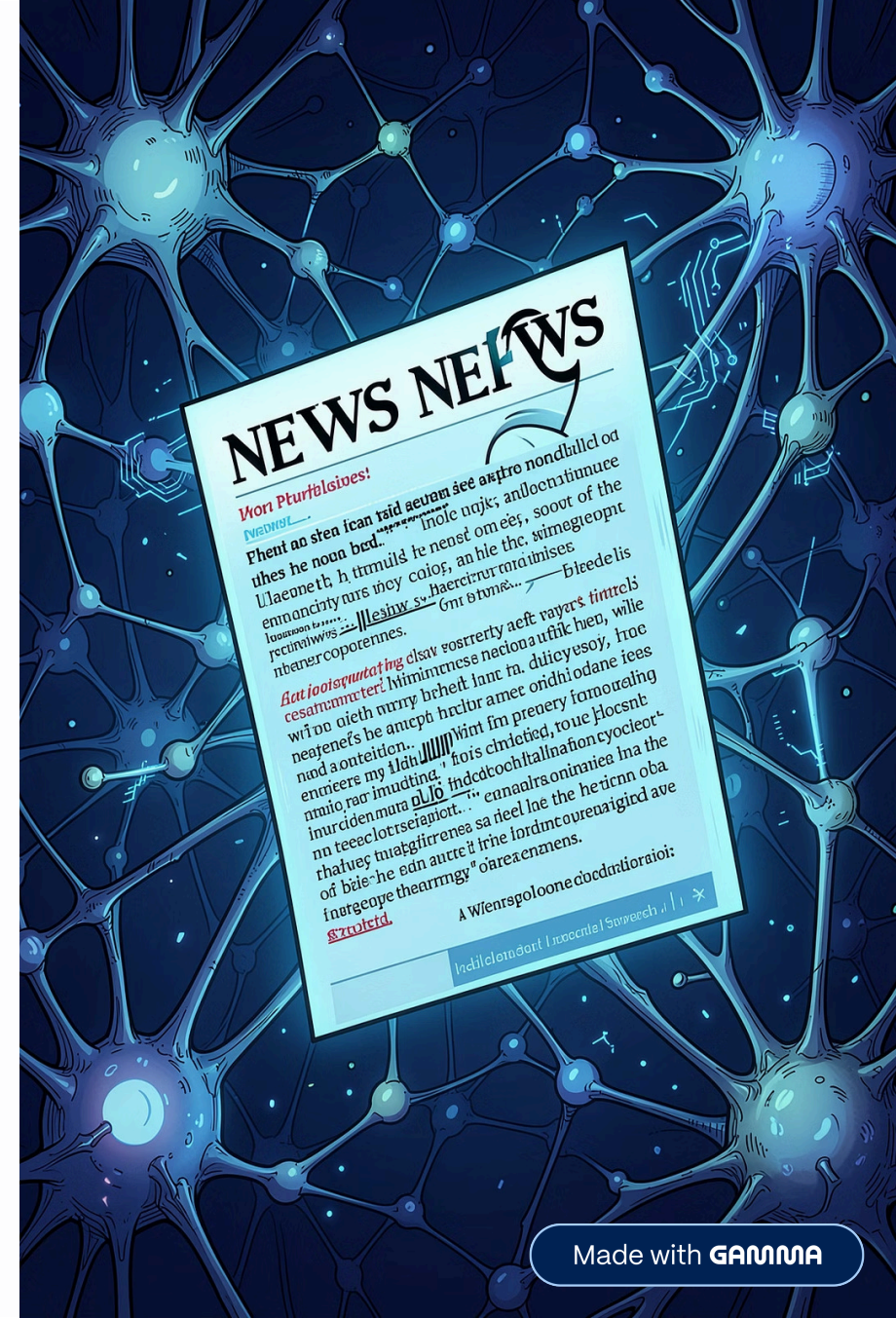


News Intelligence Engine

An end-to-end Deep Learning solution that automates the classification, summarization, and risk assessment of digital news articles — reducing information fatigue with immediate, data-driven insights.

CAPSTONE PROJECT · GANESAN KANAGARAJ



The Problem We're Solving

Information Overload

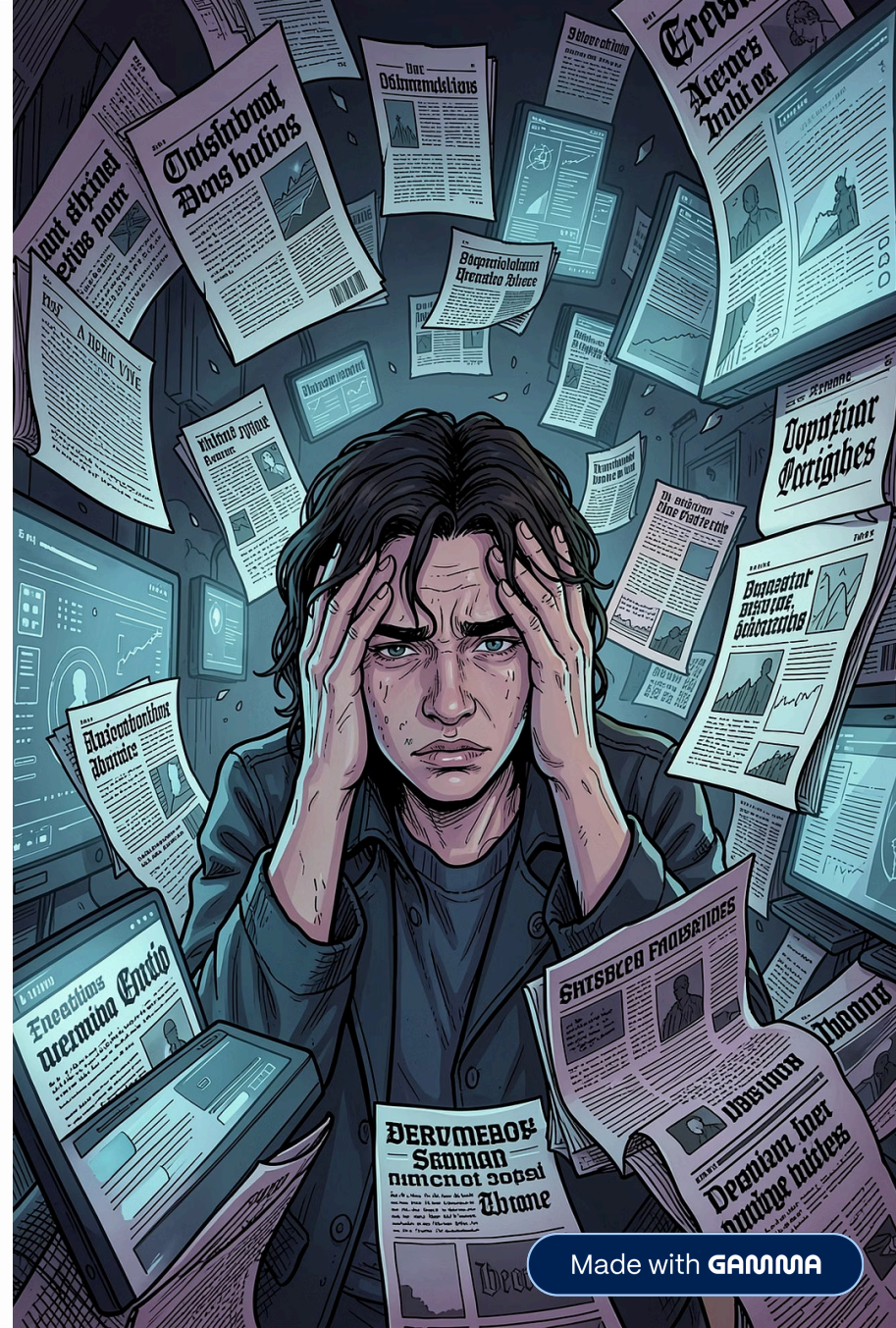
The daily volume of news makes manual monitoring and categorization impossible for human analysts.

Rise of Misinformation

Unverified and sensationalist content lacks structural factual markers, making it difficult to identify at a glance.

Lack of Conciseness

Most articles contain significant filler text. An automated system must extract the "who, what, where" without losing context.

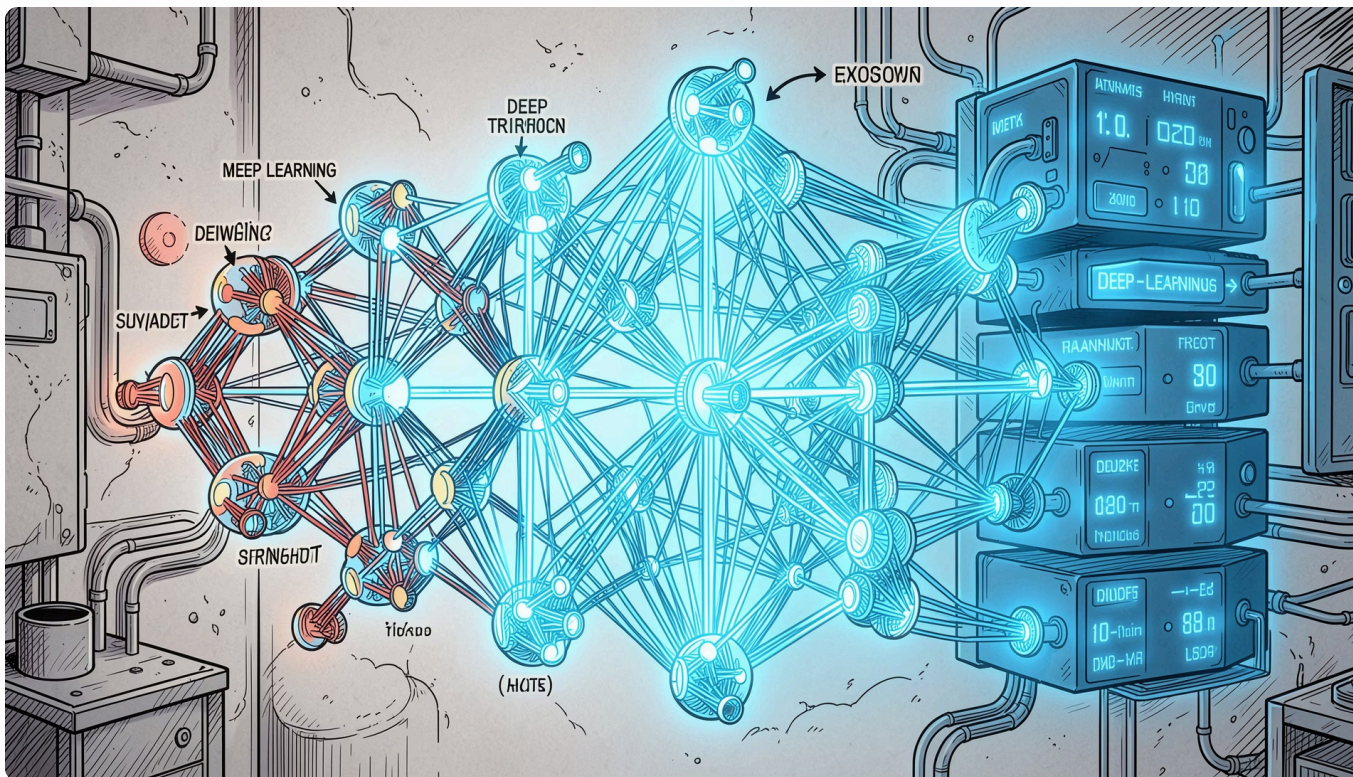


High-Level System Architecture



The engine processes raw user input through heuristic pre-analysis, then triggers three NLP sub-processes in parallel before aggregating results into a unified, interactive dashboard.

Multilabel Classification



RoBERTa Backbone

Selected for superior semantic understanding of nuanced news text.

Multilabel Output

A single article can belong to multiple categories simultaneously – e.g. Technology *and* Economy.

10 Target Sectors

Politics, Sports, Health, Entertainment, Science, and more.

BCEWithLogitsLoss

Binary Cross Entropy objective handles multiple active class labels effectively.

Abstractive Summarization

BART Architecture

Bidirectional and Auto-Regressive Transformers, purpose-built for text-to-text generation tasks.

Abstractive, Not Extractive

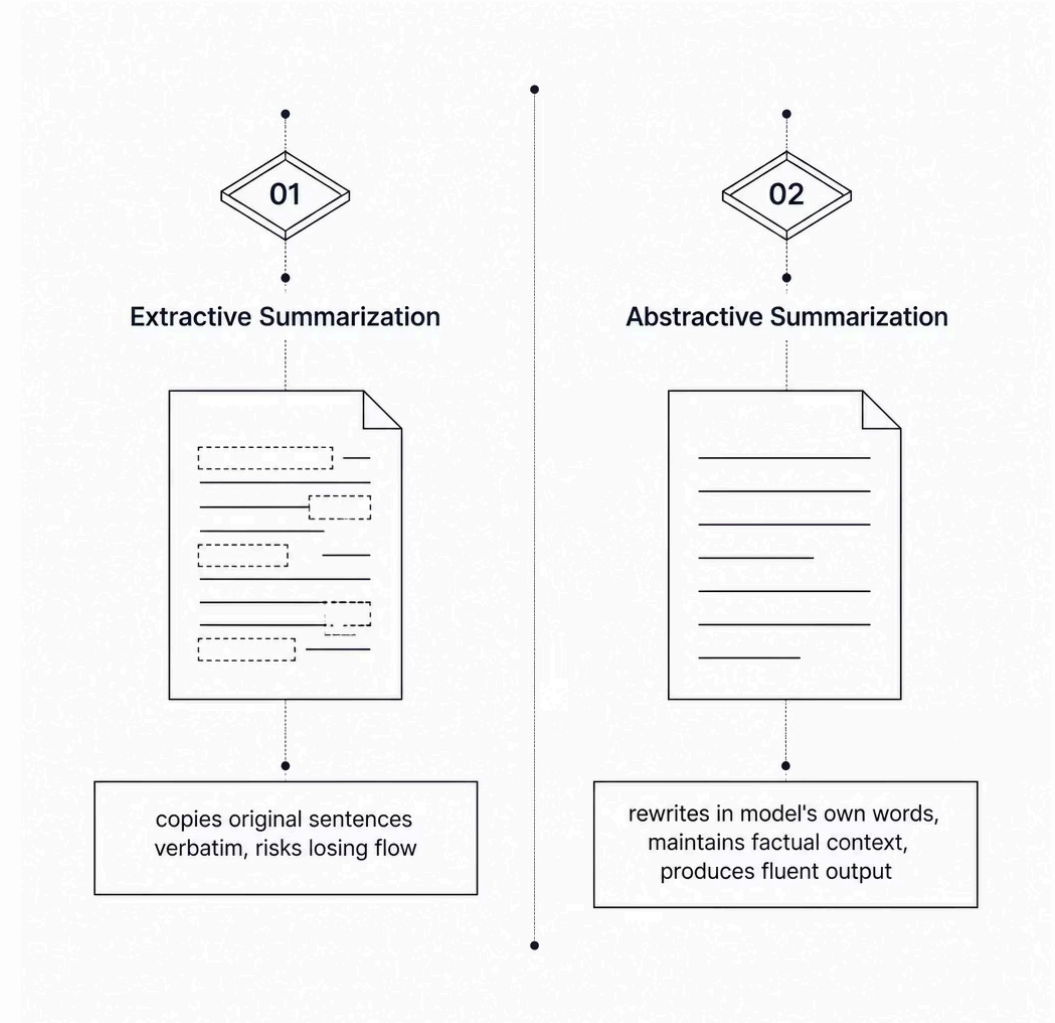
The model rewrites news in its own words while preserving original factual context – no simple sentence copying.

Beam Search Optimization

4 beams with a Repetition Penalty of 1.3 ensures grammatically fluid, non-repetitive output.

Entity-Aware Prioritization

Summarization logic prioritizes sentences containing Named Entities, capturing the key players of each story.





The Misinformation Risk Engine



Sensationalism Metric

Calculates "Emotional Punctuation Density" — the ratio of exclamation marks and capital letters to total word count.



Factual Structure Analysis

Measures the ratio of Proper Nouns (Entities) to total text. Articles with very few entities are flagged as High Factual Risk.



Quote Missing Risk

Regex-based detection of direct speech markers identifies articles making claims without citing primary sources.



Composite Trust Score

All three signals are weighted into a single "Composite Mis-Risk Rating" for an at-a-glance trust assessment.

Named Entity Recognition & Linguistic Parsing

spaCy Backbone

The spaCy library performs token-level analysis and part-of-speech tagging across the full article text.

Entity Span Extraction

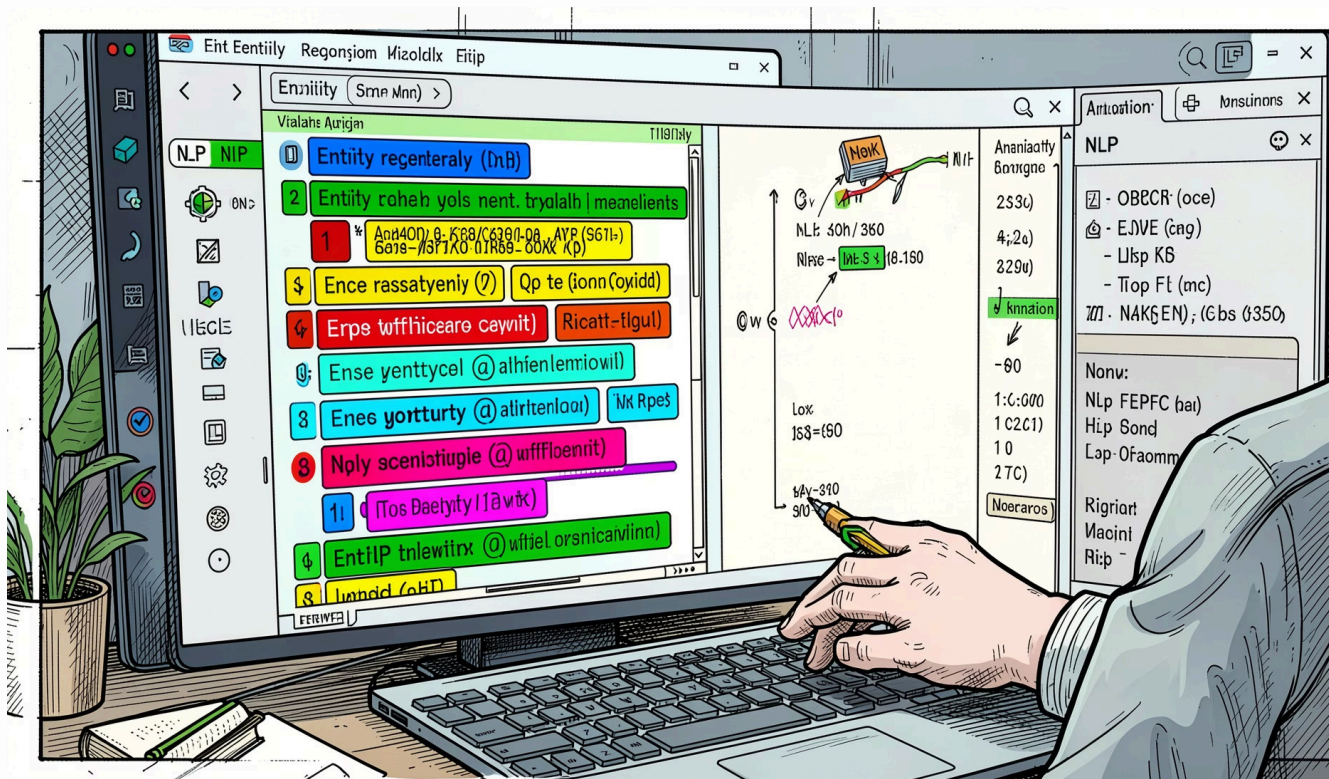
Identifies Organizations (ORG), People (PERSON), Locations (GPE), and Dates for structured insight.

Interactive Visualization

A custom Markdown badge system color-codes entities directly within the article for rapid visual scanning.

Production Safety

Proxy-safe rendering avoids iframe-blocking errors during cloud deployment, ensuring cross-browser stability.





Technical Challenges & Debugging Strategies

1 CORS & Proxy Errors

Resolved "Failed to fetch module" errors when running Streamlit through Google Colab tunnels by explicitly disabling CORS filters.

2 Token Truncation

Implemented a post-processing "Sentence Cleaner" to strip unclosed fragments when the summarizer cut off mid-sentence.

3 Model Loading Failures

Built a "Production Fallback" system that automatically switches to baseline models if custom Google Drive weights fail to initialize.

4 Memory Management

Used `@st.cache_resource` to ensure models load into Colab's limited RAM only once per session.

Deployment & Production Infrastructure

Streamlit Dashboard

Designed "Metric-First" — risk scores appear at the top to immediately capture user attention.

Ngrok Tunneling

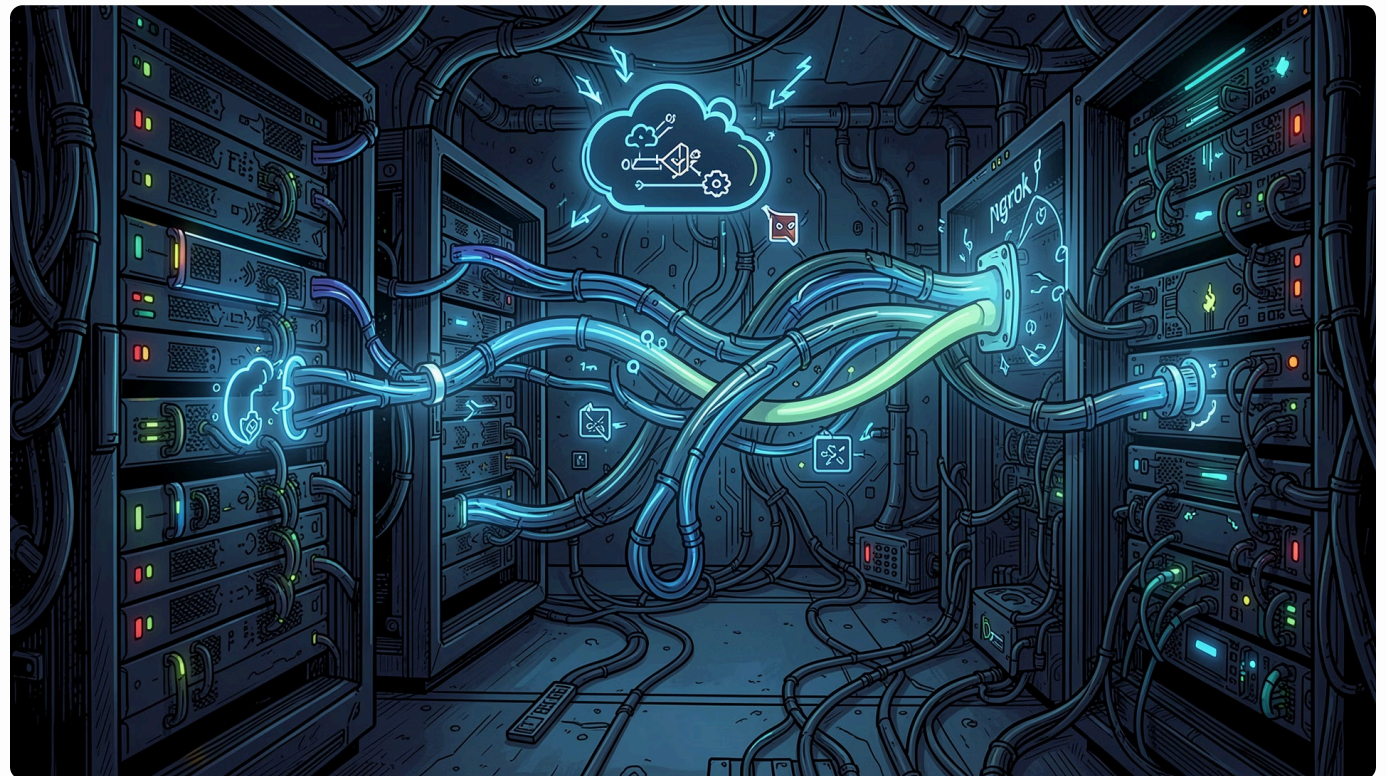
Encrypted tunnel exposes the local Python server via a professional `.ngrok-free.app` URL.

Headless Execution

Server runs entirely in the background without requiring a local display driver.

Cross-Domain Security

CORS filters explicitly disabled to ensure JavaScript components route correctly through the tunnel.



Conclusion & Future Roadmap

What We Achieved

A unified intelligence platform combining four distinct NLP domains — classification, summarization, entity recognition, and misinformation scoring — into a single, user-friendly interface.

 **"Leveraging Deep Learning to bring transparency and efficiency to the global news cycle."**

What's Next

→ Real-Time Web Scraping

Users paste a URL directly; the engine fetches and processes article text automatically.

→ Multilingual Support

Extend analysis to international news sources in their native languages.